

Question Bank and PYQs

Units and Measurement

2 Mark Questions

Q1: Define the term 'unit' in the context of physical measurement. (2 Marks)

- Measurement of any physical quantity involves comparison with a basic, arbitrarily chosen, internationally accepted reference standard called a unit.
- The result of a measurement is expressed by a numerical measure accompanied by this unit.
- A complete set of base and derived units is known as a system of units.

Q2: Distinguish between fundamental units and derived units. (2 Marks)

- The units for the fundamental or base quantities are called fundamental or base units.
- The units of all other physical quantities can be expressed as combinations of the base units.
- Such units obtained for the derived quantities are called derived units.

Q3: What is the *Système Internationale d' Unites* (SI)? (2 Marks)

- The SI is the internationally accepted system of units for measurement, abbreviated from the French name.
- It was developed by the Bureau International des Poids et Mesures (BIPM) and recently revised in 2018.
- It consists of seven base units and is widely used in scientific, technical, and commercial work.

Q5: What are significant figures in a measurement? (2 Marks)

- Significant figures are the digits in a reported measurement that include all reliably known digits plus the first digit that is uncertain.
- For example, in a period of 1.62 s, the digits 1 and 6 are certain, while 2 is uncertain.
- The number of significant figures indicates the precision of the measurement.

Q6: State the Principle of Homogeneity of Dimensions. (2 Marks)

- This principle states that the dimensions of all the terms in a physical equation must be the same.
- Magnitudes of physical quantities can only be added or subtracted if they have the same dimensions.
- If the dimensions of terms on both sides of an equation are not identical, the equation is considered wrong.

Q7: Define 'dimensional formula' with an example. (2 Marks)

- A dimensional formula is an expression showing how and which of the base quantities represent the dimensions of a physical quantity.
- For example, the dimensional formula of volume is $[M^0L^3T^0]$.
- Similarly, the dimensional formula for acceleration is $[M^0LT^{-2}]$.

Q8: What is meant by the 'order of magnitude' of a physical quantity? (2 Marks)

- When a number is expressed as $a \times 10^b$, the exponent b is called the order of magnitude.
- To find b , the number a is rounded off to 1 if $a \leq 5$ and to 10 if $5 < a \leq 10$.
- For example, the diameter of the earth (1.28×10^7 m) has an order of magnitude of 7.

Q10: Define dimensional equation. (2 Marks)

- An equation obtained by equating a physical quantity with its dimensional formula is called a dimensional equation.
- It represents the dimensions of the quantity in terms of base quantities.
- For example, the dimensional equation for force is $[F] = [MLT^{-2}]$.

3 Mark Questions

Q4: Define the radian and steradian as supplementary SI units. (3 Marks)

- The unit for plane angle is the radian (*rad*), defined as the ratio of arc length ds to radius r , so $d\theta = ds/r$.

- The unit for solid angle is the steradian (sr), defined as the ratio of intercepted area dA to the square of radius r^2 , so $d\Omega = dA/r^2$.
- Both plane and solid angles are considered dimensionless quantities in the SI system.

Q9: Why does a mere change of units not change the number of significant figures? (3 Marks)

- Significant figures represent the precision of measurement, which depends on the least count of the instrument used.
- A choice of different units is just a mathematical conversion and does not alter the inherent precision of the original measurement.
- For instance, 2.308 cm (four sig figs) is the same as 0.02308 m (four sig figs).

Q11: Briefly describe the CGS, FPS, and MKS systems of units. (3 Marks)

- In earlier times, scientists used three major systems of units extensively.
- In the CGS system, the base units for length, mass, and time were centimetre, gram, and second respectively.
- In the FPS system (British system), the base units were foot, pound, and second respectively.
- In the MKS system, the base units were metre, kilogram, and second respectively.
- These systems were replaced by the internationally standardized SI system.

Q12: Explain the rules for determining significant figures for numbers involving zeros. (3 Marks)

- Zeros between two non-zero digits are always significant, regardless of the decimal point.
- In numbers less than 1, zeros to the right of the decimal but to the left of the first non-zero digit are not significant.
- Terminal or trailing zeros in a number without a decimal point are not significant.
- Trailing zeros in a number with a decimal point are significant as they indicate precision.
- The zero put to the left of a decimal for a number less than 1 is never significant.

Q13: State the rules for arithmetic operations (multiplication and division) with significant figures. (3 Marks)

- In multiplication or division, the final result should retain as many significant figures as the original number with the least significant figures.
- For example, if density is calculated as $4.237 \text{ g}/2.51 \text{ cm}^3$, the result must be reported to three significant figures (1.69 g cm^{-3}).
- This ensures the result is not more precise than the least precise input data.
- Intermediate results should retain one extra digit to avoid cumulative rounding errors.

Q14: Describe the rounding-off conventions used in scientific measurements. (3 Marks)

- If the digit to be dropped is more than 5, the preceding digit is raised by 1.
- If the digit to be dropped is less than 5, the preceding digit is left unchanged.
- If the digit to be dropped is exactly 5, the preceding digit is left unchanged if it is even.
- If the digit to be dropped is exactly 5 and the preceding digit is odd, it is raised by 1.
- For example, 2.745 rounds to 2.74, while 2.735 rounds to 2.74.

Q15: Determine the dimensions of Force and Work using base quantities. (3 Marks)

- Force is the product of mass and acceleration.
- Acceleration has dimensions $\text{length}/\text{time}^2 = [LT^{-2}]$.
- Therefore, dimensions of Force are $[M] \times [LT^{-2}] = [MLT^{-2}]$.
- Work is defined as Force $\times$ displacement, so its unit is $J = \text{kg m}^2\text{s}^{-2}$.
- Thus, dimensions of Work/Energy are $[MLT^{-2}] \times [L] = [ML^2T^{-2}]$.

Q16: Check the dimensional consistency of the equation $\frac{1}{2}mv^2 = mgh$. (3 Marks)

- The dimensions of the LHS term ($\frac{1}{2}mv^2$) are $[M][LT^{-1}]^2 = [ML^2T^{-2}]$.
- The constant $\frac{1}{2}$ is dimensionless.
- The dimensions of the RHS term (mgh) are $[M][LT^{-2}][L] = [ML^2T^{-2}]$.
- Since dimensions of LHS = dimensions of RHS, the equation is dimensionally correct.
- Dimensional consistency does not guarantee the numerical correctness of constants.

Q17: Explain the importance of scientific notation in reporting measurements. (3 Marks)

- Scientific notation expresses a number as $a \times 10^b$, where $1 \leq a < 10$.
- It removes ambiguity regarding the significance of trailing zeros.
- For example, 4.700 m written as 4.700×10^3 mm clearly shows four significant figures.
- The power of 10 is irrelevant for determining significant figures, which reside only in the base number a .
- It simplifies numerical computations for very large or small physical quantities.

Q18: Why is it necessary to specifying a standard for comparison when calling a quantity 'large' or 'small'? (3 Marks)

- Calling a dimensional quantity 'large' or 'small' is meaningless without a standard reference.
- For example, saying 'atoms are very small' is meaningful only if compared to a macroscopic object like a tip of a pin.
- Similarly, the mass of Jupiter is 'large' compared to Earth but 'small' compared to the Sun.
- A jet plane moves with 'great speed' relative to a car but 'slowly' relative to the speed of light.
- Measurements in physics rely on the comparison of a quantity with an established unit.

Q19: Each side of a cube is 7.203 m. Calculate its surface area to appropriate significant figures. (3 Marks)

- The measured length 7.203 m has four significant figures.
- The formula for surface area of a cube is $6l^2$.
- Calculation: $6 \times (7.203)^2 = 311.299254 \text{ m}^2$.
- Since the input length has 4 sig figs, the result must be rounded to 4 sig figs.
- The surface area is reported as 311.3 m^2 .

Q20: Explain relative error using an example from the text. (3 Marks)

- Relative error is the ratio of absolute error to the measured value, often expressed as a percentage.
- For a mass of 1.02 g with an accuracy of ± 0.01 g, the relative error is $(\pm 0.01/1.02) \times 100\% \approx \pm 1\%$.
- For a mass of 9.89 g with the same accuracy, the relative error is $(\pm 0.01/9.89) \times 100\% \approx \pm 0.1\%$.
- This shows that relative error depends on both the accuracy and the magnitude of the measured quantity.
- Precision is better reflected by relative error than by absolute error alone.

5 Mark Questions

Q21: List and define the seven SI base quantities and their units. (5 Marks)

- 1. Metre (m): The unit of length, defined by the speed of light in vacuum $c = 299792458 \text{ m s}^{-1}$.
- 2. Kilogram (kg): The unit of mass, defined by the Planck constant $h = 6.62607015 \times 10^{-34} \text{ J s}$.
- 3. Second (s): The unit of time, defined by the caesium frequency $\Delta VC s = 9192631770 \text{ Hz}$.
- 4. Ampere (A): The unit of electric current, defined by the elementary charge $e = 1.602176634 \times 10^{-19} \text{ C}$.
- 5. Kelvin (K): The unit of thermodynamic temperature, defined by the Boltzmann constant $k = 1.380649 \times 10^{-23} \text{ J K}^{-1}$.
- 6. Mole (mol): The unit for amount of substance, containing exactly $6.02214076 \times 10^{23}$ elementary entities.
- 7. Candela (cd): The unit of luminous intensity, defined by the luminous efficacy of monochromatic radiation.
- These definitions are based on progress in technology to ensure greater precision and accuracy.

Q22: Explain the applications and limitations of dimensional analysis in physics. (5 Marks)

- Application 1: Checking dimensional consistency of equations. An equation is incorrect if dimensions of all terms are not identical.
- Application 2: Deducing relations among physical quantities by assuming a product-type dependence on base quantities.
- Application 3: Unit conversion. It allows checking if units are treated like algebraic symbols to ensure correct derivation.
- Limitation 1: It cannot determine the value of dimensionless constants like 2π or $1/2$.
- Limitation 2: A dimensionally correct equation may not be physically exact or correct.
- Limitation 3: It cannot distinguish between different physical quantities having the same dimensions, like work and torque.
- Limitation 4: It cannot handle complex dependencies involving trigonometric or logarithmic functions that must be dimensionless.
- It serves as a preliminary test of consistency when the correctness of an equation is in doubt.

Q23: Provide a detailed explanation of the rules for determining significant figures in measurements. (5 Marks)

- Significant figures consist of reliable digits plus the first uncertain digit in a measurement.
- Rule 1: All non-zero digits are significant (e.g., 287.5 has four).
- Rule 2: Zeros between non-zero digits are significant, regardless of the decimal point.
- Rule 3: Leading zeros in numbers less than 1 are not significant (e.g., in 0.002308, the underlined zeros are not significant).
- Rule 4: Trailing zeros in a number without a decimal are not significant (e.g., 123000 has three sig figs).
- Rule 5: Trailing zeros in a number with a decimal are significant (e.g., 3.500 has four).
- Scientific notation ($a \times 10^b$) removes ambiguity; all zeros in the base number a are significant.
- Multiplying or dividing factors that are exact numbers have infinite significant figures (e.g., 2 in $2\pi r$).

Q24: Derive the expression for the time period (T) of a simple pendulum using the method of dimensions. (5 Marks)

- Assume the time period T depends on the length of the string (l), mass of the bob (m), and acceleration due to gravity (g).
- Express the relation as a product: $T = k \cdot l^x \cdot g^y \cdot m^z$, where k is a dimensionless constant.
- Substitute the dimensions: $[L^0 M^0 T^1] = [L]^x [LT^{-2}]^y [M]^z$.
- Combine base dimensions: $[L^0 M^0 T^1] = L^{x+y} T^{-2y} M^z$.
- Equate exponents of corresponding dimensions on both sides: $x + y = 0$, $-2y = 1$, and $z = 0$.
- Solve the linear equations: $y = -1/2$, $x = 1/2$, and $z = 0$.
- Substitute these values back: $T = k \cdot l^{1/2} \cdot g^{-1/2} = k\sqrt{l/g}$.
- Note that dimensional analysis cannot determine that $k = 2\pi$.

Q25: Explain the rules for addition and subtraction of significant figures with examples. (5 Marks)



- Rule: The final result of an addition or subtraction should retain as many decimal places as the number with the least decimal places.
- Example 1: Add 436.32 g, 227.2 g, and 0.301 g. Arithmetic sum is 663.821 g.
- The least precise measurement (227.2 g) has one decimal place, so the final result is 663.8 g.
- Example 2: Subtract 0.304 m from 0.307 m. The result is 0.003 m.
- Writing this in scientific notation gives 3×10^{-3} m, which maintains the precision.
- Unlike multiplication, the number of significant figures may change; the focus is on decimal places.
- For instance, $12.9 \text{ g} - 7.06 \text{ g}$ results in 5.8 g (one decimal place), even if inputs have three sig figs each.
- This ensures the precision of the result properly conveys the uncertainty of the measurements.

Q26: Elaborate on how dimensions are used in mechanics with examples of volume and force. (5 Marks)

- The nature of any physical quantity is defined by its dimensions, represented by powers of base quantities.
- In mechanics, the base quantities are Mass $[M]$, Length $[L]$, and Time $[T]$.
- Volume is the product of three lengths: length $\times$ breadth $\times$ height $= [L] \times [L] \times [L] = [L^3]$.
- Since volume does not depend on mass or time, it has dimensions $[M^0 L^3 T^0]$.
- Force is the product of mass and acceleration.
- Acceleration is change in velocity over time, which is (length/time)/time $= [LT^{-2}]$.
- The dimensions of Force are $[M] \times [LT^{-2}] = [MLT^{-2}]$.
- Magnitudes do not matter in dimensional representation; only the quality of the type of quantity enters.

Q27: Discuss the rules for reporting results in multi-step calculations to maintain precision. (5 Marks)

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- Final results should not have more significant figures than the original data.
 - Intermediate results should retain one more digit than the number of significant digits in the least precise measurement.
 - Rounding off should only occur at the end of the calculation to avoid building up rounding errors.
 - Example: Calculating the reciprocal of 9.58. Rounded to three sig figs, $1/9.58 = 0.104$.
 - However, the reciprocal of 0.104 rounded to three digits is 9.62, which does not retrieve the original value.
 - If we retain an extra digit ($1/9.58 = 0.1044$), the reciprocal returns 9.58 correctly.
 - This justifies why the extra digit is essential during intermediate multi-step calculations.
 - Consistency in units and algebraic treatment of units must be maintained throughout all steps.

Q28: Describe the SI supplementary units and differentiate them from base units. (5 Marks)

- Supplementary units were defined for angles in addition to the seven base units.
- Radian (rad) is the unit for plane angle $d\theta = ds/r$, where ds is arc length and r is radius.
- Steradian (sr) is the unit for solid angle $d\Omega = dA/r^2$, where dA is spherical area and r is radius.
- Refer to [Image of plane angle and solid angle description] for geometric clarity.
- Base units (like metre or kilogram) represent physical quantities that have dimensions $[L]$ or $[M]$.
- Supplementary units are dimensionless because they are ratios of identical physical quantities (length/length or area/length squared).
- Unlike base units, which are defined by fundamental constants (like h or c), these are defined geometrically.
- They are included in the SI system to ensure completeness in scientific and technical work.

Q29: Explain how dimensional analysis helps in checking the consistency of the kinematic equation $x = x_0 + v_0t + \frac{1}{2}at^2$. (5 Marks)

- The equation represents distance x traveled by a body starting at x_0 with initial velocity v_0 and acceleration a .
- Identify the dimensions of each term independently.
- 1. Term on LHS: $[x] = [L]$.
- 2. First term on RHS: $[x_0] = [L]$.
- 3. Second term on RHS: $[v_0t] = [LT^{-1}] \times [T] = [L]$.
- 4. Third term on RHS: $[\frac{1}{2}at^2] = [LT^{-2}] \times [T^2] = [L]$.
- The numerical factor $\frac{1}{2}$ is dimensionless and does not affect the analysis.
- Since all terms have the same dimension $[L]$, the equation is dimensionally consistent.

Q30: How are units of common quantities like volume, density, and speed derived from SI base units? (5 Marks)

- Derived units are combinations of the seven base units (m, kg, s, A, K, mol, cd).

- Volume is length $\times$ length $\times$ length, so its SI unit is $m \times m \times m = m^3$.
- Density is mass/volume, so its derived SI unit is $kg/m^3 = kg\ m^{-3}$.
- Speed is distance/time, leading to the derived SI unit $m/s = m\ s^{-1}$.
- Acceleration is change in speed over time, derived as $(ms^{-1})/s = m\ s^{-2}$.
- Energy (Joule) is derived from work as $kg\ m^2s^{-2}$.
- Table 1.2 shows that some units like litre ($10^{-3}m^3$) are retained for general use though they are outside the standard SI.
- Using SI prefixes and multiples helps in expressing these derived units for very large or small measurements.

